

Smart Healthcare Analytics And Patient Insights

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1. Project Introduction

1.1 Background and Motivation

Healthcare organizations generate large amounts of patient-related data daily, including patient demographics, diseases, medications, treatment costs, insurance claims, and hospital outcomes. However, converting this raw healthcare data into meaningful insights for operational and strategic decision-making remains a major challenge.

With increasing healthcare demands and rising treatment costs, hospitals require intelligent analytics systems that can provide real-time insights into patient care and hospital performance. This project leverages SAP Analytics Cloud (SAC) to develop an interactive healthcare analytics solution that enables hospitals to monitor patient trends, treatment efficiency, financial performance, and healthcare outcomes through advanced visualizations and dashboards.

1.2 Problem Statement

Hospitals and healthcare institutions often face difficulties in:

- Tracking patient admission and discharge patterns
- Identifying high-risk diseases and costly treatments
- Monitoring patient recovery and treatment outcomes
- Managing insurance claims effectively
- Generating centralized reports for decision-making
- Analyzing regional healthcare trends

The goal of this project is to build a dashboard that addresses these challenges and enables data-driven decisions.

1.3 Objectives of the Project

- To analyze patient demographics and disease distribution
- To monitor patient admission and discharge trends
- To evaluate treatment costs and insurance claim patterns
- To measure patient recovery and healthcare outcomes
- To build interactive and dynamic dashboards using SAC
- To provide actionable insights for hospital management

1.4 Project Setup

<https://www.sap.com/products/data-cloud/cloud-analytics/trial.html>

The screenshot shows the SAP Analytics Cloud trial page. At the top is the SAP navigation bar with links for Products, Industries, Transform and Support, Learning, Community, Partners, and About. Below this is a breadcrumb trail: / SAP Business Data Cloud / SAP Analytics Cloud. The main heading is "SAP Analytics Cloud Trial". A message states: "Visit our local site for more information and offerings for your country." with buttons for "SAP India" and "Stay on the United States site". The primary offer is "SAP Analytics Cloud advanced trial" with a prominent blue button labeled "Start your free advanced trial". Below this is a purple banner for "SAP Sapphire" with the text "Discover how SAP's AI, data, and platform technologies improve operations, performance, and innovation." and a "Register now" button. Further down, the text "Try SAP Analytics Cloud for free" is followed by a paragraph: "Start your digital transformation today for 60 days with the SAP Analytics Cloud advanced trial. Connect your data sources, model your data, create visualizations, add augmented analytics, and try predictive planning." A "Contact us" button is located at the bottom right.

Click "Start Your Free Advanced Trial."

This screenshot shows the same SAP Analytics Cloud trial page as before, but with a modal form open in the center. The modal is titled "Access SAP Trial" and contains the text "Please provide your e-mail to get started". It features a text input field labeled "Business e-mail address *" with the email "gurnulesnehal151@gmail.com" entered. A blue "Continue" button is positioned at the bottom of the modal. The background page content is dimmed, showing the "Start your free advanced trial" button and the "Try SAP Analytics Cloud" section.

Enter your SAP email address.

Check your email inbox for the verification link.



System & Provisioning

Tenant Access Information for SAP Analytics Cloud

Thank you for choosing SAP Analytics Cloud. Please find below your account details to access the application.

Tenant URL <https://sactrial-sacap10-r8ytjtdz051tvtt7zvcjqg7u.ap10.hcs.cloud.sap>

Username: gurnulesnehal151@gmail.com

Password If you cannot remember your password, please click the "Forgot Password?" link on the login page. You will receive a separate email with instructions on how to set a new password. After setting your password, you can log on as described above.

You have received the tenant URL. Please use it to log in.

SAP HANA Cloud Services

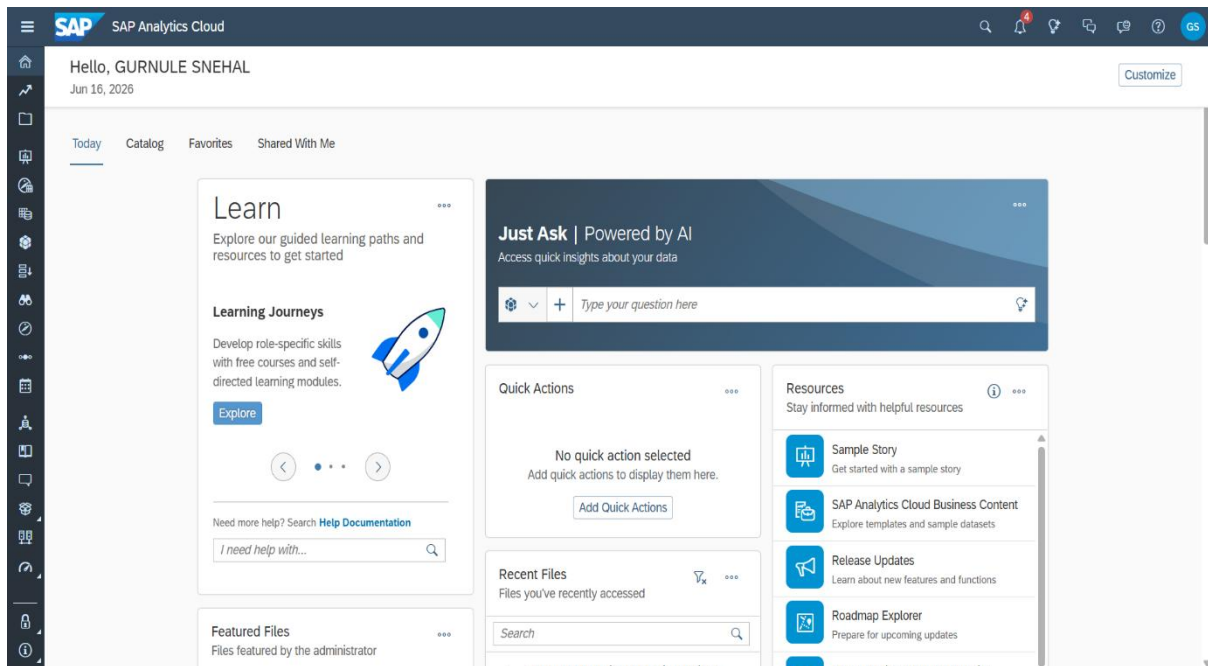
Sign In

Email or User Name

Password

☐ Keep me signed in [Forgot password?](#)

[Continue](#)



This is your SAP Analytics Cloud system.

2. Dataset and Data Preprocessing

2.1 Data Source

The dataset used for this project is a hospital patient data records dataset collected from Kaggle containing patient treatment and hospital-related information.

	z2 Patient_ID	z2 Age	AA Gender	AA Disease	AA Medication	Admission...	Discharge...	AA Patient_S...	AA Outcome	AA Insurance...
1	1	45	Female	Heart Disease	Angioplasty	2024-02-07	2024-02-12	Kerala	Recovered	Yes
2	2	60	Male	Diabetes	Insulin Therapy	2023-03-11	2023-03-14	Goa	Stable	No
3	3	32	Female	Fractured Arm	X-Ray and Splint	2025-01-02	2025-01-03	Maharashtra	Recovered	Yes
4	4	75	Male	Stroke	CT Scan and Mec	2023-12-30	2024-01-06	Manipur	Stable	No
5	5	50	Female	Cancer	Surgery and Cher	2022-05-10	2022-05-20	Mizoram	Recovered	No
6	6	68	Male	Hypertension	Medication and C	2025-06-27	2025-06-29	Sikkim	Stable	Yes
7	7	55	Female	Appendicitis	Appendectomy	2023-12-13	2023-12-17	Karnataka	Recovered	No
8	8	40	Male	Fractured Leg	Cast and Physical	2022-07-28	2022-08-03	Gujarat	Recovered	Yes
9	9	70	Female	Heart Attack	Cardiac Catheteri	2023-05-11	2023-05-19	Kerala	Stable	No
10	10	25	Male	Allergic Reaction	Epinephrine Injec	2023-04-25	2023-04-26	Kerala	Recovered	No
11	11	48	Female	Respiratory Infect	Antibiotics and R	2024-03-28	2024-03-30	Delhi	Stable	No
12	12	65	Male	Prostate Cancer	Radiation Therapy	2025-05-05	2025-05-14	Delhi	Recovered	No
13	13	30	Female	Childbirth	Delivery and Post	2025-10-04	2025-10-07	Bihar	Recovered	No
14	14	52	Male	Kidney Stones	Lithotripsy	2025-09-24	2025-09-28	Delhi	Recovered	No
15	15	58	Female	Osteoarthritis	Physical Therapy	2023-10-28	2023-11-02	Chandigarh	Stable	No
16	16	55	Female	Heart Disease	Angioplasty	2022-10-27	2022-11-02	Nagaland	Recovered	Yes
17	17	62	Male	Diabetes	Insulin Therapy	2022-07-04	2022-07-08	Goa	Stable	No
18	18	35	Female	Fractured Arm	X-Ray and Splint	2025-10-25	2025-10-27	Maharashtra	Recovered	Yes

2.2 Dataset Attributes

The dataset contains the following fields:

- Patient_ID
- Age
- Gender
- Medication
- Admission_Date
- Discharge_Date
- Disease
- Patient_State
- Outcome
- Insurance_Claimed
- Treatment_Cost

2.3 Data Cleaning and Transformation

The following preprocessing steps were performed:

- Removed null and duplicate records
- Converted date fields into standard date format
- Standardized disease and medication names
- Verified treatment cost values
- Cleaned inconsistent outcome records
- Prepared structured dataset for analytics and visualization

This step ensured clean and structured data for SAC.

The screenshot shows the SAP Data Builder interface. The main table displays 18 rows of patient data with columns: Patient_ID, Age, Gender, Disease, Medication, Admission_Date, and Discharge_Date. The right-hand 'Dataset Overview' pane shows the initial configuration with 2 measures (Age, Treatment_Cost) and 9 dimensions (Patient_ID, Gender, Disease, Medication, Admission_Date, Discharge_Date, Patient_State, Outcome, Insurance_Claimed).

Patient_ID	Age	Gender	Disease	Medication	Admission_Date	Discharge_Date
1	45	Female	Heart Disease	Angioplasty	2024-02-07	2024-02-12
2	60	Male	Diabetes	Insulin Therapy	2023-03-11	2023-03-14
3	32	Female	Fractured Arm	X-Ray and Splint	2025-01-02	2025-01-03
4	75	Male	Stroke	CT Scan and Med	2023-12-30	2024-01-06
5	50	Female	Cancer	Surgery and Cher	2022-05-10	2022-05-20
6	68	Male	Hypertension	Medication and C	2025-06-27	2025-06-29
7	55	Female	Appendicitis	Appendectomy	2023-12-13	2023-12-17
8	40	Male	Fractured Leg	Cast and Physical	2022-07-28	2022-08-03
9	70	Female	Heart Attack	Cardiac Catheteri	2023-05-11	2023-05-19
10	25	Male	Allergic Reaction	Epinephrine Injec	2023-04-25	2023-04-26
11	48	Female	Respiratory Infect	Antibiotics and Re	2024-03-28	2024-03-30
12	65	Male	Prostate Cancer	Radiation Therap	2025-05-05	2025-05-14
13	30	Female	Childbirth	Delivery and Post	2025-10-04	2025-10-07
14	52	Male	Kidney Stones	Lithotripsy	2025-09-24	2025-09-28
15	58	Female	Osteoarthritis	Physical Therapy	2023-10-28	2023-11-02
16	55	Female	Heart Disease	Angioplasty	2022-10-27	2022-11-02
17	62	Male	Diabetes	Insulin Therapy	2022-07-04	2022-07-08
18	35	Female	Fractured Arm	X-Ray and Splint	2025-10-25	2025-10-27

The screenshot shows the same dataset after a transformation. A new column, 'Insurance_Claimed', has been added, with values of 1 for patients 1, 2, 3, 6, 8, 16, 17, and 18, and 0 for all other patients. The 'Dataset Overview' pane on the right shows the updated configuration with 3 measures (Age, Treatment_Cost, Insurance_Claimed) and 8 dimensions (Gender, Disease, Medication, Admission_Date, Discharge_Date, Patient_State, Outcome).

Patient_ID	Age	Gender	Disease	Medication	Admission_Date	Discharge_Date	Insurance_Claimed
1	45	Female	Heart Disease	Angioplasty	2024-02-07	2024-02-12	1
2	60	Male	Diabetes	Insulin Therapy	2023-03-11	2023-03-14	1
3	32	Female	Fractured Arm	X-Ray and Splint	2025-01-02	2025-01-03	1
4	75	Male	Stroke	CT Scan and Med	2023-12-30	2024-01-06	0
5	50	Female	Cancer	Surgery and Cher	2022-05-10	2022-05-20	0
6	68	Male	Hypertension	Medication and C	2025-06-27	2025-06-29	1
7	55	Female	Appendicitis	Appendectomy	2023-12-13	2023-12-17	0
8	40	Male	Fractured Leg	Cast and Physical	2022-07-28	2022-08-03	1
9	70	Female	Heart Attack	Cardiac Catheteri	2023-05-11	2023-05-19	0
10	25	Male	Allergic Reaction	Epinephrine Injec	2023-04-25	2023-04-26	0
11	48	Female	Respiratory Infect	Antibiotics and Re	2024-03-28	2024-03-30	0
12	65	Male	Prostate Cancer	Radiation Therap	2025-05-05	2025-05-14	0
13	30	Female	Childbirth	Delivery and Post	2025-10-04	2025-10-07	0
14	52	Male	Kidney Stones	Lithotripsy	2025-09-24	2025-09-28	0
15	58	Female	Osteoarthritis	Physical Therapy	2023-10-28	2023-11-02	0
16	55	Female	Heart Disease	Angioplasty	2022-10-27	2022-11-02	1
17	62	Male	Diabetes	Insulin Therapy	2022-07-04	2022-07-08	1
18	35	Female	Fractured Arm	X-Ray and Splint	2025-10-25	2025-10-27	1

Fig :Transformed Insurance_Claimed column in dataset

2.4 Derived Metrics and Calculations

Additional calculated fields were created:

- Recovery Rate = (Recovered Patients / Total Patients) × 100
- Average Treatment Cost = Total Treatment Cost / Total Patients

3. SAP Analytics Cloud (SAC) Implementation

3.1 Data Import in SAC

The cleaned healthcare dataset was imported into SAP Analytics Cloud using the Import Model feature

Measures:

- Treatment_Cost
- Insurance_Claimed
- Age

Dimensions:

- Disease
- Gender
- Patient_State
- Outcome
- Medication

Date Dimension:

- Admission_Date
- Discharge_Date

3.2 Data Modeling and Preparation

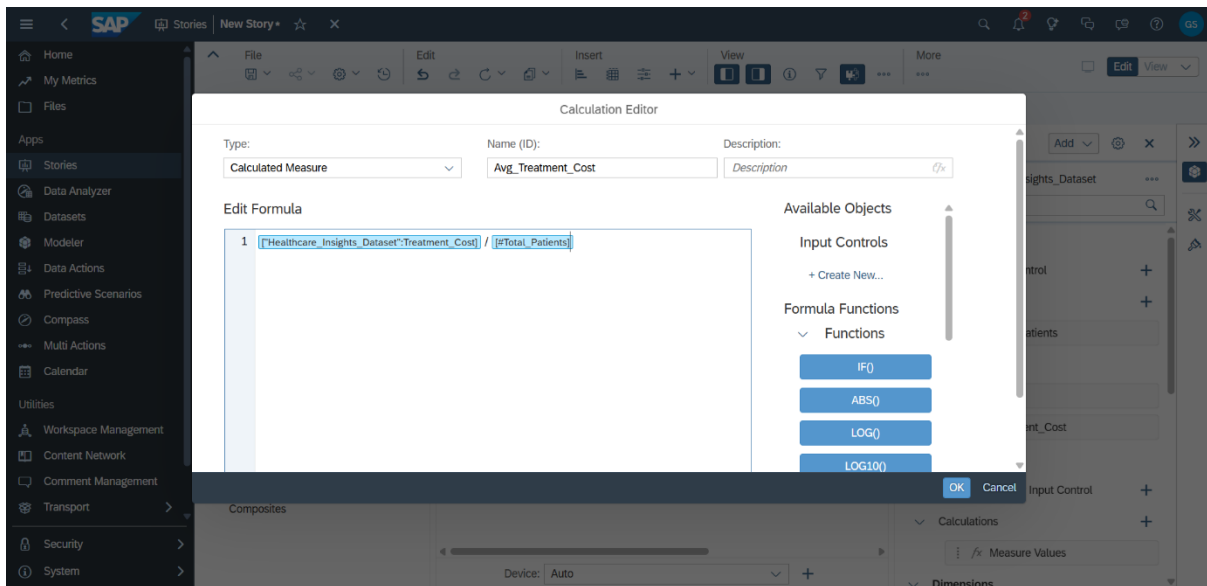
The imported data was modelled using dimensions, measures, hierarchies, and calculated fields to support dynamic analysis and storytelling capabilities in SAC.

3.3 Calculated Measures and KPIs

The following KPIs were created:

- Total Patients
- Recovery Rate
- Average Treatment Cost

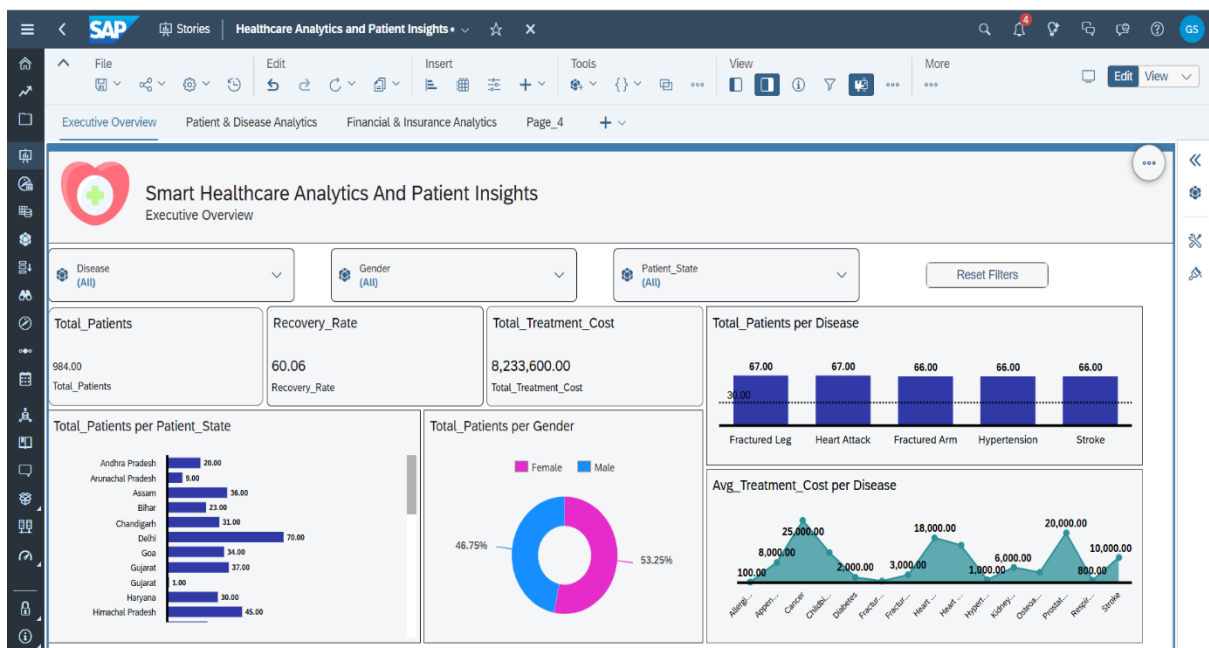
These KPIs are used in dashboard visualization.



4. Dashboard Design and Development

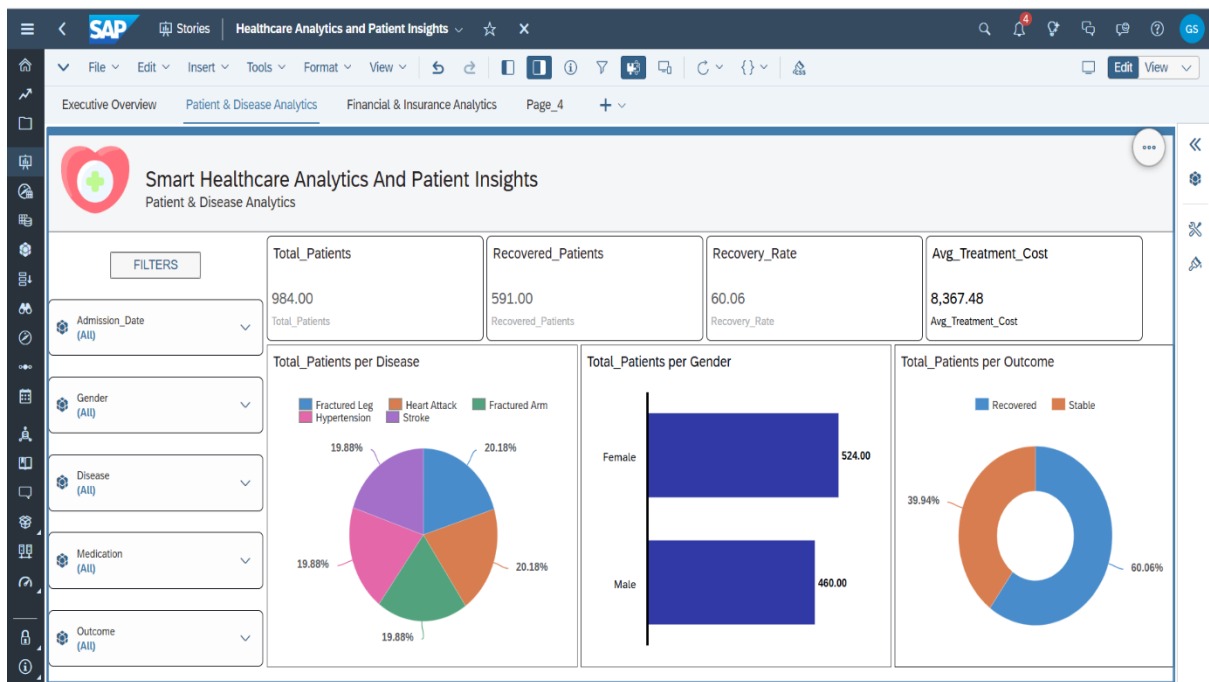
4.1 Executive Overview Page

- Donut Chart:
 - Total Patients per Gender
- Bar Chart:
 - Total Patients per State



4.2 Patient and Disease Analytics

- Pie Chart:
 - Total Patients per Disease
- Bar Chart:
 - Total Patients per Gender
- Donut Chart:
 - Total Patients per Outcome



4.3 Financial and Insurance Analytics

- Bar Chart:
 - Treatment Cost per Disease
- Tree Map:
 - Insurance Claimed per Disease
- Trend Chart:
 - Avg_Treatment_Cost per Disease

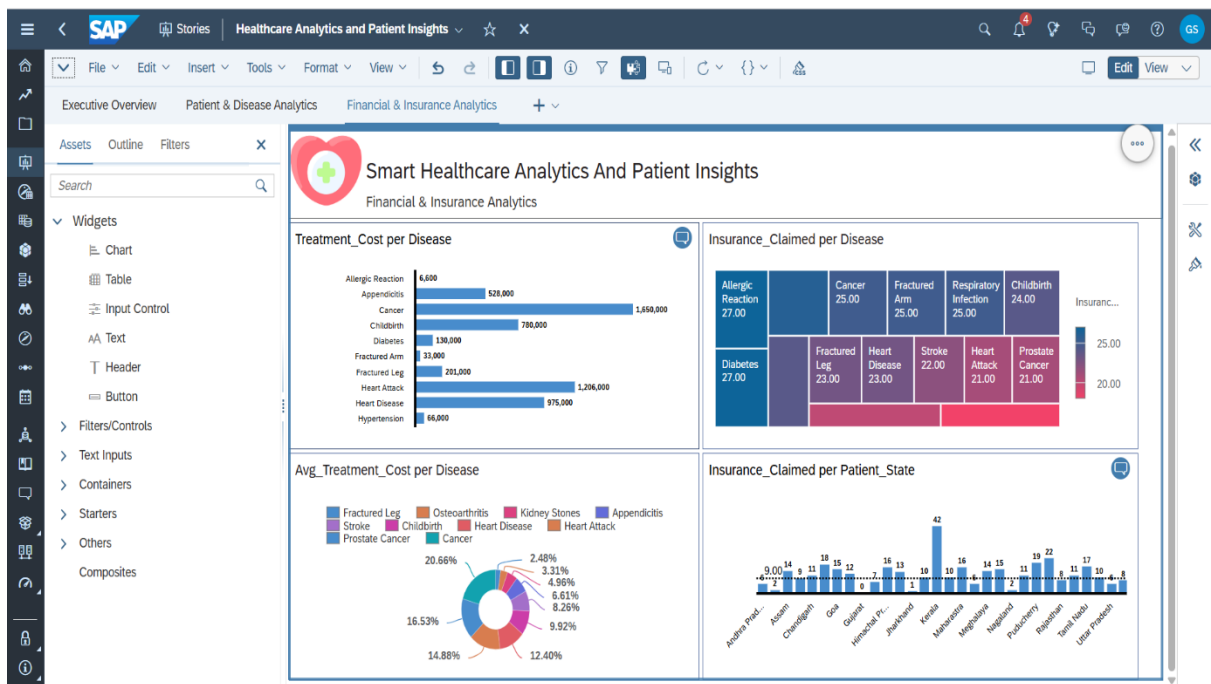
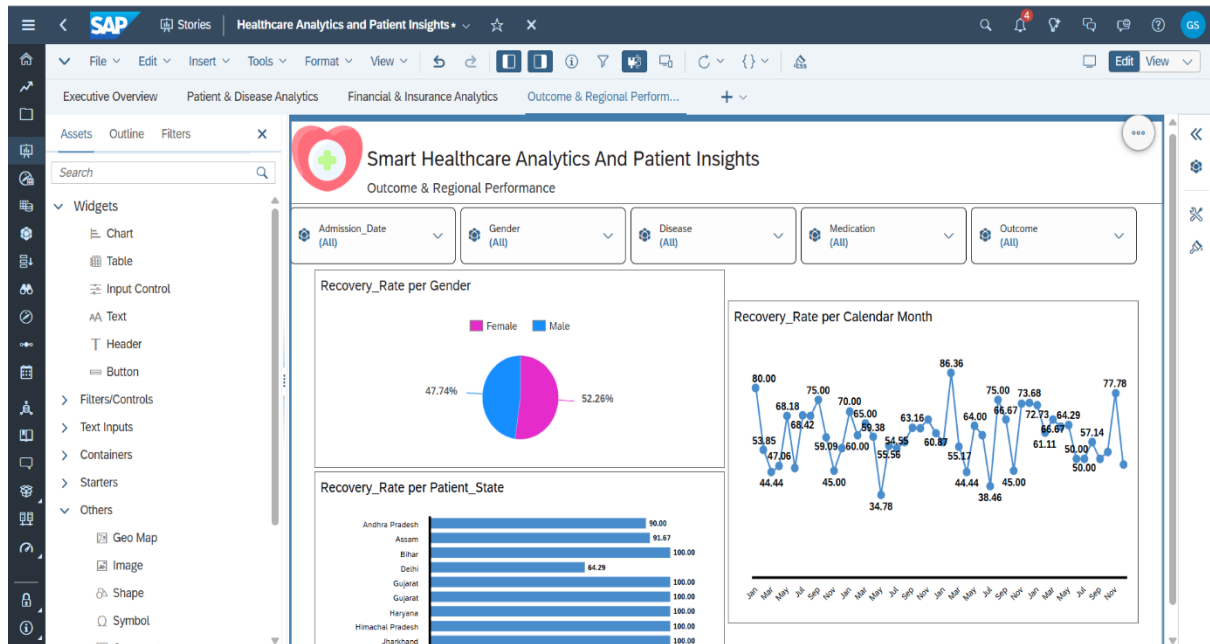


Fig: Widget Commenting

4.4 Outcome and Regional Performance

- Patient Outcome Distribution
- Recovery Trends Over Time
- Regional Healthcare Performance Analysis



5. Visualization and Features

5.1 Charts and Graphs Used

- Line Chart → Trends
- Bar Chart → Comparisons
- Pie Chart → Distribution
- Tree Map → Contribution
- Donut Chart → Proportion & Percentages of Whole

5.2 Filters and Input Controls

Implemented filters:

- Disease
- Gender
- Outcome
- Patient State
- Admission Date
- Medication

These filters allow dynamic exploration of data.

5.3 Smart Insights and Forecasting

- Detect disease patterns
- Identify high-cost treatments
- Forecast future patient admissions
- Analyze healthcare trends

6. Results, Insights and Business Recommendations

6.1 Patient and Disease Insights

- Certain diseases showed significantly higher patient occurrence
- Recovery rates varied across diseases and age groups
- Some medications were associated with longer recovery periods

6.2 Financial and Insurance Insights

- Critical diseases generated higher treatment costs
- Insurance claims were higher for specialized treatments
- Some states showed greater healthcare expenditure

6.3 Regional Healthcare Trends

- Urban regions reported higher patient admissions
- Certain states experienced seasonal disease patterns
- Regional healthcare demand varied significantly

6.4 Recommendations for Hospitals

- Improve focus on high-risk diseases
- Optimize resource allocation across departments
- Enhance insurance claim processing efficiency
- Use predictive analytics for hospital planning

7. Presentation and Demonstration

7.1 Story Navigation and Interactivity

- Multi-page story navigation
- Dynamic filtering and drill-down analysis
- Real-time interaction between charts and KPIs

7.2 Dashboard Explanation and Communication

- Clear explanation of healthcare KPIs
- Logical analytical storytelling
- Professional and user-friendly visual design

8. Conclusion

The project successfully demonstrates how SAP Analytics Cloud can transform healthcare data into meaningful insights through interactive dashboards and analytics. By analyzing patient trends, treatment costs, insurance claims, and healthcare outcomes, the solution supports smarter healthcare management, operational efficiency, and data-driven decision-making.

9. Future Enhancements

- Real-time hospital database integration
- AI-based disease prediction models
- Advanced patient risk analysis
- Mobile healthcare analytics dashboards
- Integration with Electronic Health Record (EHR) systems

10. References

- Kaggle Healthcare Dataset
- SAP Analytics Cloud Documentation
- SAP Learning Resources
- Healthcare Analytics Research Articles